

CHAPTER 3: DESIGN TRADEOFFS

This chapter focuses on the evaluation of three architectural designs based on established engineering constraints through a trade-off analysis. The three designs Monolithic, Decoupled, and Enterprise are evaluated to determine which architecture most effectively satisfies the salon's operational needs. The analysis results from the computation of five specific design constraints: Usability (User Satisfaction Score), Functionality (Task Success Rate), Performance (Page Load Speed), Reliability (System Availability), and Maintainability (Code Simplicity Score). The values of each constraint are normalized to enable an objective comparison and determine the most suitable design for the project. Additionally, a sensitivity analysis is performed to verify the stability of the winning design under varying stakeholder priorities.

3.1 Summary of Constraints

The three designs are assessed by comparing them against these constraints to determine their efficacy in resolving the problem of scheduling and satisfying the project objectives. The table below lists the raw metrics for each design alternative, covering the five primary engineering constraints established for the Bléum Nails By Micah Appointment System

Table 3.1 Summary of Design Constraints

Designs	Constraints				
	Usability (Scale 1-10)	Functionality (Task Success %)	Performance (Page Load Sec)	Reliability (Availability %)	Maintainability (Complexity 1-10)
Design 1: Monolithic	7	80%	2.5s	99.5%	3
Design 2: Decoupled	10	95%	1.1s	98.0%	6
Design 3: Enterprise	6	98%	0.7s	99.9%	9

The design constraints highlight the tradeoffs among the three proposed architectures based on their User Satisfaction Score, Task Success Rate, Page Load Speed, System Availability, and Code Simplicity Score.

3.2 Trade-offs

The trade-off method is an approach used by engineers to assess and compare different designs based on their performance characteristics, or by integrating overall performance across multiple criteria (Otto & Antonsson, 1991). It establishes constraints that are prioritized according to their relative importance and stakeholder preferences.

In this project, three design alternatives were evaluated against five constraints using the Pareto Multi-Criteria Decision-Making approach. Each design was rated based on the simulation conducted per constraints, reflecting how well it meets the project objectives.

Table 3.2 shows the design constraints, indicating whether each should be minimized or maximized, along with their assigned raw importance and corresponding percentage values derived from the converted data.

Table 3.2 Preference and Importance of Constraints

Constraints	Preference	Importance (raw)	% Importance
Usability	Maximization	10	25.0%
Functionality	Maximization	9	22.5%
Performance	Minimization	8	20.0%
Reliability	Maximization	7	17.5%
Maintainability	Minimization	6	15.0%

To obtain a normalized value for minimization, subtract the raw value from the highest value in the dataset. Divide the result by the range (the difference between the highest and lowest values). Then, multiply this quotient by nine and add one. This process is expressed in Equation 3.1.

$$\text{Minimization} = 9 \times \left(\frac{\text{Max Value} - \text{Raw Value}}{\text{Max Value} - \text{Min Value}} \right) + 1 \text{ Equation No. 3.1}$$

To obtain a normalized value for maximization, the difference between the highest value and the raw value must be obtained. The quotient is then multiplied by nine and then added to 1. The formula is Equation 3.2 to calculate the normalized value for maximization.

$$\text{Maximization} = 9 \times \left(\frac{\text{Raw Value} - \text{Min Value}}{\text{Max Value} - \text{Min Value}} \right) + 1 \text{ Equation No. 3.2}$$

3.2.1 Tradeoff 1: Usability

The first constraint evaluated is usability, where the preference value is based on the highest satisfaction score. Design 2 (Decoupled) has the highest usability score of 10, while the lowest score is 6. After applying Equation 3.2, the Monolithic design yielded a normalized value of 3.25, the Decoupled design yielded 10.00, and the Enterprise design yielded 1.00.

Table 3.2.1 Evaluation of Three Design Alternatives based on Usability

Design	Usability
Design 1: Monolithic	7
Design 2: Decoupled	10
Design 3: Enterprise	6

3.2.1.1 Design 1: Normalization of Monolithic (Usability)

The first constraint to be evaluated is the usability. In this constraint, the preference value is based on the highest usability score. Design 2 has the highest usability score of 10, while the lowest score is 6.

$$Maximization = 9 \times \left(\frac{7-6}{10-6} \right) + 1$$

After the computation, the Monolithic design yielded a normalized value of **3.25**.

3.2.1.2 Design 2: Normalization of Decoupled (Usability)

In terms of usability, Design 2 has the highest score at 10, while the lowest value is 6. Since this constraint follows maximization, the formula applied is equation 3.2.

$$Maximization = 9 \times \left(\frac{10-6}{10-6} \right) + 1$$

After the computation, the Decoupled design yielded a normalized value of **10.00**.

3.2.1.3 Design 3: Normalization of Enterprise (Usability)

Design C has a usability score of 6, which lies between the highest and lowest values. Since the objective is to maximize usability, the maximization formula is applied.

$$Maximization = 9 \times \left(\frac{6-6}{10-6} \right) + 1$$

After the computation, the Enterprise design yielded a normalized value of **1.00**.

3.2.2 Tradeoff 2: Functionality

This constraint measures the Task Success Rate per design. Design 3 (Enterprise) has the highest functionality at 98%, while Design 1 (Monolithic) is recorded at 80%. Using Equation 3.2, the Monolithic design yielded a normalized value of 1.00, the Decoupled design yielded 8.50, and the Enterprise design yielded 10.00.

Table 3.2.2 Evaluation of Three Design Alternatives based on Functionality

Design	Functionality
Design 1: Monolithic	80%
Design 2: Decoupled	95%
Design 3: Enterprise	98%

3.2.2.1 Design 1: Normalization of Monolithic (Functionality)

The second constraint to be evaluated is the functionality. In this constraint, the preference value is based on the highest functionality score. The highest functionality score is 98%, while the lowest score is 80%.

$$Maximization = 9 \times \left(\frac{80-80}{98-80} \right) + 1$$

After the computation, the Monolithic design yielded a normalized value of **1.00**.

3.2.2.2 Design 2: Normalization of Decoupled (Functionality)

In terms of functionality, Design 3 has the highest score at 98%, while the lowest value is 80%. Since this constraint follows maximization, the formula applied is equation 3.2.

$$Maximization = 9 \times \left(\frac{95-80}{98-80} \right) + 1$$

After the computation, the Decoupled design yielded a normalized value of **8.50**.

3.2.2.3 Design 3: Normalization of Enterprise (Functionality)

In terms of functionality, Design 3 has the highest score at 98%, while the lowest value is 80%. Since this constraint follows maximization, the formula applied is equation 3.2.

$$Maximization = 9 \times \left(\frac{98-80}{98-80} \right) + 1$$

After the computation, the Enterprise design yielded a normalized value of **10.00**.

3.2.3 Tradeoff 3: Performance

Performance is evaluated by page load speed in seconds. Design 3 (Enterprise) has the fastest performance at 0.7s, while Design 1 (Monolithic) is the slowest at 2.5s. Following the minimization formula (Eq. 3.1), the Monolithic design yielded a normalized value of 1.00, the Decoupled design yielded 8.00, and the Enterprise design yielded 10.00.

Table 3.2.3 Evaluation of Three Design Alternatives based on Performance

Design	Performance
Design 1: Monolithic	2.5s
Design 2: Decoupled	1.1s
Design 3: Enterprise	0.7s

3.2.3.1 Design 1: Normalization of Monolithic (Performance)

The third constraint to be evaluated is performance. In this constraint, the preference value is based on the highest performance score. The highest performance score is 0.7s, while the lowest score is 2.5s.

$$\text{Minimization} = 9 \times \left(\frac{2.5-2.5}{2.5-0.7} \right) + 1$$

After the computation, the Monolithic design yielded a normalized value of **1.00**.

3.2.3.2 Design 2: Normalization of Decoupled (Performance)

In terms of performance, Design 2 has a score of 1.1s, while the highest value is 0.7s and the lowest value is 2.5s. Since this constraint follows minimization, the formula applied is equation 3.1.

$$\text{Minimization} = 9 \times \left(\frac{2.5-1.1}{2.5-0.7} \right) + 1$$

After the computation, the Decoupled design yielded a normalized value of **8.00**.

3.2.3.3 Design 3: Normalization of Enterprise (Performance)

Design 3 has a performance score of 0.7s, which is the highest value. Since the objective is to minimize performance, the minimization formula is applied.

$$\text{Minimization} = 9 \times \left(\frac{2.5-0.7}{2.5-0.7} \right) + 1$$

After the computation, the Enterprise design yielded a normalized value of **10.00**.

3.2.4 Tradeoff 4: Reliability

Reliability is identified by system availability. Design 3 has the highest reliability score at 99.9%, while Design 2 is at 98.0%. Using Equation 3.2, the Monolithic design yielded a normalized value of 8.11, the Decoupled design yielded 1.00, and the Enterprise design yielded 10.00.

Table 3.2.4 Evaluation of Three Design Alternatives based on Reliability

Design	Reliability
Design 1: Monolithic	99.5%
Design 2: Decoupled	98.0%
Design 3: Enterprise	99.9%

3.2.4.1 Design 1: Normalization of Decoupled (Reliability)

In terms of reliability, Design 1 has a score of 99.5%, while the highest value is 99.9% and the lowest value is 98.0%. Since this constraint follows maximization, the formula applied is equation 3.2.

$$Maximization = 9 \times \left(\frac{99.5 - 98}{99.9 - 98} \right) + 1$$

After the computation, the Monolithic design yielded a normalized value of **8.11**.

3.2.4.2 Design 2: Normalization of Decoupled (Reliability)

In terms of reliability, Design 2 has a score of 98.0% which is the lowest value, while the highest value is 99.9%. Since this constraint follows maximization, the formula applied is equation 3.2.

$$Maximization = 9 \times \left(\frac{98 - 98}{99.9 - 98} \right) + 1$$

After the computation, the Decoupled design yielded a normalized value of **1.00**.

3.2.4.3 Design 3: Normalization of Enterprise (Reliability)

Design 3 has a reliability score of 99.9%, which is the highest value. Since the objective is to maximize reliability, the maximization formula is applied.

$$Maximization = 9 \times \left(\frac{99.9 - 98}{99.9 - 98} \right) + 1$$

After the computation, the Enterprise design yielded a normalized value of **10.00**.

3.2.5 Tradeoff 5: Maintainability

The final constraint evaluated is maintainability, the preference value is based on the lowest complexity score, where a lower score indicates simpler code. Design 1 has the highest code simplicity 3, while Design 3 is the most complex at 9. Applying Eq 3.1, the Monolithic design yielded a normalized value of 10.00, the Decoupled design yielded 5.50, and the Enterprise yielded 1.00.

Table 3.2.5 Evaluation of Three Design Alternatives based on Maintainability

Design	Maintainability
Design 1: Monolithic	3
Design 2: Decoupled	6
Design 3: Enterprise	9

3.2.5.1 Design 1: Normalization of Monolithic (Maintainability)

The last constraint to be evaluated is maintainability. In this constraint, the preference value is based on the lowest maintainability score. The highest maintainability score is 9, while the lowest score is 3. Since this constraint follows minimization, the formula is applied.

$$\text{Minimization} = 9 \times \left(\frac{9-3}{9-3} \right) + 1$$

After the computation, the Monolithic design yielded a normalized value of **10.00**.

3.2.5.2 Design 2: Normalization of Decoupled (Maintainability)

In terms of maintainability, Design 2 has a score of 6, which is the middle value, while the lowest value is 3. Since this constraint follows minimization, the formula is applied.

$$\text{Minimization} = 9 \times \left(\frac{9-6}{9-3} \right) + 1$$

After the computation, the Decoupled design yielded a normalized value of **5.50**.

3.2.5.3 Design 3: Normalization of Enterprise (Maintainability)

Design 3 has a maintainability score of 9, which is also the highest value. Since the objective is to minimize maintainability, the minimization formula is applied.

$$\text{Minimization} = 9 \times \left(\frac{9-9}{9-3} \right) + 1$$

After the computation, the Enterprise design yielded a normalized value of **1.00**.

3.3 Summary of the Normalized Values of the Three Designs

The table below summarizes the normalized values derived from the mathematical evaluation of the three architectural designs

Table 3.3 Normalized Value of the Three Designs

Designs	Constraints				
	Usability	Functionality	Performance	Reliability	Maintainability
Design 1: Monolithic	3.25	1.00	1.00	8.11	10.00
Design 2: Decoupled	10.00	8.50	8.00	1.00	5.50
Design 3: Enterprise	1.00	10.00	10.00	10.00	1.00

3.4 Designers Raw Ranking for the Three Designs

The final ranking is determined by multiplying each normalized score by its corresponding percentage importance and summing the results.

Table 3.4 Designers Raw Ranking for the Three Designs

Decision Criteria	Criterion's Importance		Ability to Satisfy Criterion		
	Scale (0-10)	Percentage (%)	Design 1: Monolithic	Design 2: Decoupled	Design 3: Enterprise
Usability	10	25.0%	0.81	2.50	0.25
Functionality	9	22.5%	0.23	1.91	2.25
Performance	8	20.0%	0.20	1.60	2.00
Reliability	7	17.5%	1.42	0.18	1.75
Maintainability	6	15.0%	1.50	0.83	0.15
TOTAL SCORE	40	100%	4.16	7.02	6.40

3.5 Sensitivity Analysis

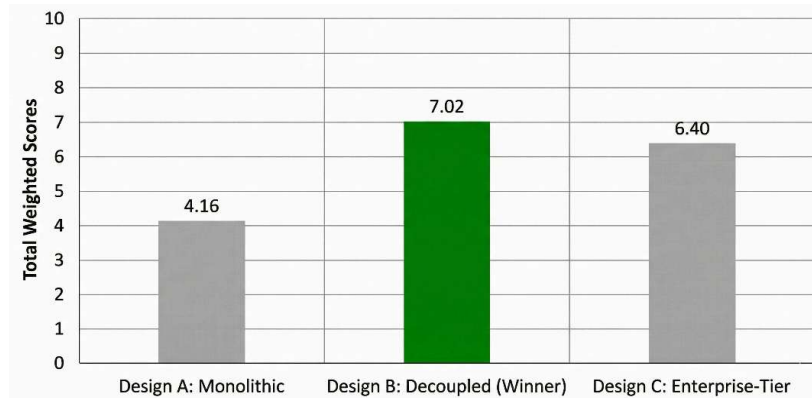


Figure 3.1 Sensitivity Analysis Using Main Bar Chart

The sensitivity analysis evaluates the stability of the design ranking by simulating changes in the importance weights of the established constraints. If the project's priority shifted toward ease of maintenance due to limited human resources, such as in a solo developer context, the weight for Maintainability would significantly increase. In this specific simulation, the Monolithic architecture of Design A would emerge as the superior choice because its single codebase structure provides a level of simplicity that outweighs the user experience advantages of the other designs. Conversely, if the business model expanded into a large-scale franchise requiring the handling of thousands of concurrent users, the weight for Performance and Reliability would be prioritized. Under these enterprise conditions, Design C would become the optimal choice due to the multi-threading capabilities of the Java ecosystem and its mission critical stability. Ultimately, the current analysis confirms that Design B is a highly stable winner for a contemporary, client-facing salon, maintaining its lead as long as Usability and Functionality weights remain above 25%. It only loses ground in extreme edge case scenarios where technical simplicity or massive enterprise scale performance are prioritized over the actual end-user experience.

3.6 Influence of the Design Tradeoffs in the Final Design

The mathematical evaluation identified Design B, the Decoupled Full-Stack Architecture, as the winning design with a total score of 7.02, which directly influenced the final technical implementation of the Bléum Nails By Micah system. This high Usability score reinforced the decision to implement a JavaScript-based Single Page Application (SPA), ensuring the fluid, app-like transitions required by the salon's demographic. Furthermore, the priority placed on Functionality led to the selection of a Python (FastAPI), which allows for the asynchronous execution of the Capacity-Limiting Algorithm to enforce daily client limits in real-time without blocking the user interface. The trade-off between reliability and maintainability led to the utilization of a MySQL Relational Database, providing the necessary structure to secure sensitive client allergy records and appointment history while remaining more manageable for the development team. Finally, selecting Design B enabled the integration of secure QRPh localized payments and Role-Based Access Control (RBAC) in compliance with Republic Act No. 10173, offering a professional level of security without the verbose code complexity of an enterprise-tier setup.